DEPARTMENT OF curriculum +instruction | STEM education

Timing

1. You are eligible to take the written exam when (a) you are in at least the 4th semester of the program, (b) your advisor(s) approves, (c) your program plan has been submitted and approved (d) you have completed the following coursework: (i) STEM Core Courses: CI 8571, CI 8572, CI 8573, (ii) Research Methods: CI 8133, and at least one of CI 8148, EdPsy 8261, 8262, 8251, or 8252, and (iii) Focus Area Courses: MTHE 8571 or 1 science education course, or 1 engineering course
2. A 3-week window is identified each semester for the written exam
3. You will have 10 calendar days to complete the written exam
4. You will receive written feedback summarizing the readers' comments within two weeks
5. After you complete the written exam, consult with your advisor to schedule the preliminary oral exam. Full-time students must complete the preliminary oral within one year of completing the written exams and part-time students within two years

Written Exam

The written exam is a qualifying exam to determine if you have the depth of knowledge and writing ability necessary to successfully complete the program. You will be evaluated on your ability to:

- synthesize ideas and show independent thinking
- write clearly
- set up an argument for a position
- demonstrate skill in synthesizing historical and current research to build a case in response to a question
- demonstrate in written form substantive understanding of foundational knowledge connected to Ph.D. coursework.

Take-home exam, consisting of two questions:

1. assessing your understanding of coursework related to the STEM core courses and research methods courses. This question will ask a

Oral Exam

The purpose of the preliminary oral exam is to determine readiness for successful completion of the dissertation.

The exam will assess:

- depth and understanding of your knowledge of STEM and focus area content
- knowledge of research methodologies
- ability to engage in oral arguments and discourse

The oral exam is forward looking, explores possible research topics and designs for future studies.

student to either critique a research report or paper addressing strengths and limitations in connection to its theoretical framework, methods and conclusions or to develop a research proposal based on a critical issue within STEM Education.

2. assessing foundational knowledge from the three STEM core courses and focus area coursework.

Three STEM faculty will form your written exam committee. Exam readers include your advisor, another faculty member from your focus area and a STEM Education faculty member from outside your focus area.

The chair of the committee will notify you about the exam results.

Possible results are: pass, partial pass (rewrites required), and fail.

If you fail the written exam, you may take the exam one more time. The retake exam may not be taken in the same semester as the original exam but must be completed within a 12-month period of the original exam. If you fail to retake exam you will be removed from the PhD program.

You should be able to identify research questions related to critical issues in their field, select research designs and methods related to the research question, and discuss different paradigms of research that could be employed.

Committee members will ask questions related to the written exams.

In addition, the supporting program committee member will have the opportunity to ask questions related to the supporting area coursework.

You will be asked to share current thoughts about you dissertation, however, the prelim exam does not include any formal evaluation of the dissertation proposal.